POWER PULSE

Digital Meters Important Formulae

Notes by Balaji Sir

1 Resolution

1. $$\text{Resolution} = \frac{1}{10^{n}} \ (n \text{ full digits}), \qquad 3\tfrac{1}{2} \text{ digit: max } 1999, \text{ range } 2\text{ V} \Rightarrow 1\text{ mV}$$

a half digit can display only 0 or 1

2 ADC and DAC (n bits)

2. $$\text{Resolution (LSB)} = \frac{V_{FS}}{2^{n}}, \qquad \text{Quantization error} = \pm\frac{1}{2}\,\text{LSB}$$

3. $$V_{out} = V_{ref}\sum_{i=1}^{n}\frac{b_i}{2^{i}}$$

DAC output; b_i = bit i (MSB first)

4. Flash ADC: $2^{n}-1$ comparators (fastest); Successive approx.: n clock cycles; Counter type: up to 2^{n}

speed ranking: flash > successive approximation > counter > dual slope

3 Dual-slope integrating DVM

5. $$V_{in} = V_{ref}\frac{T_2}{T_1}$$

T_1 = fixed integration time; T_2 = de-integration time, proportional to V_in; independent of the integrator R and C

6. $$T_1 = k \times \frac{1}{f_{noise}} \ (\text{e.g. } 20\text{ ms for } 50\text{ Hz})$$

rejects power-line interference

4 Digital frequency / period meter

7. $$f = \frac{N}{T_{gate}}, \qquad T = N\,T_{clock}, \qquad \text{error} = \pm 1 \text{ count}$$

use the frequency mode at high frequency and the period mode at low frequency